

In this chapter you will learn:

- some rules for dealing with exponents
- about a function where the unknown is in the exponent
- about the value e and some of its properties
- how to undo exponential functions using an operation called a logarithm
- the rules of logarithms
- about graphs of logarithms
- to use logarithms to find exact solutions to simple exponential equations.

2 Exponents and logarithms

Introductory problem

A radioactive substance has a half-life of 72 years. A 1 kg block of the substance is found to have a radioactivity of 25 million Becquerel (Bq). How long, to the nearest 10 years, would it take for the radioactivity to have fallen below 10 000 Bq?

Many mathematical models (biological, physical and financial in particular) involve the concept of continuous growth or decay where the rate of growth/decay of the population is linked to the size of that population. You may have met similar situations already, for example when a bank account earns compound interest: the increase in amount each year depends upon how much is in the account. Any similar situation is governed by an exponential function, which you will learn about in this chapter.

We shall also look at how we can find out how long an exponential process has been occurring, using a function called a logarithm.

2A Laws of exponents

The exponent of a number shows you how many times the number is to be multiplied by itself. You will have already met some of the rules for dealing with exponents before, and in this section we shall revisit and extend these rules. (In other courses, the exponent might have been called the 'index' or 'power'.)

See Prior learning
Section C on the CD-
ROM which looks at
exponents.



A number written in **exponent form** is one which explicitly looks like:

a^n n is referred to as the **exponent** or **power**
 a is referred to as the **base**

a^n is pronounced 'a to the exponent n' or, more simply, 'a to the n'.

To investigate the rules of exponents let us consider an example:

Worked example 2.1

Simplify.

- (a) $a^3 \times a^4$ (b) $a^3 \div a^4$ (c) $(a^4)^3$ (d) $a^4 + a^3$

$$(a) \ a^3 \times a^4 = (a \times a \times a) \times (a \times a \times a) = a^7$$

$$(b) \ a^3 \div a^4 = \frac{\cancel{a} \times \cancel{a} \times a}{a \times \cancel{a} \times \cancel{a} \times a} = \frac{1}{a} = a^{-1}$$

$$(c) \ (a^4)^3 = a^4 \times a^4 \times a^4 = a^{12}$$

$$(d) \ a^4 + a^3 = a^3(a+1)$$

Use the idea from part (a)

The example above suggests some rules of exponents.

KEY POINT 2.1

$$a^m \times a^n = a^{m+n}$$

KEY POINT 2.2

$$a^m \div a^n = a^{m-n}$$

KEY POINT 2.3

$$(a^m)^n = a^{m \times n}$$

We can use Key point 2.3 to justify the interpretation of $a^{\frac{1}{n}}$ as the n th root of a , since $(a^{\frac{1}{n}})^n = a^{\frac{1}{n} \times n} = a$. This is exactly the property we require of the n th root of a . So, we get the rule:

$$a^{\frac{m}{n}} = \left(a^{\frac{1}{n}}\right)^m.$$



Mathematics is often considered a subject without ambiguity. However, the value of 0^0 is undetermined; it depends upon how you get there!



It is questionable whether in part (d) we have actually simplified the expression. Sometimes the way mathematicians choose to simplify expressions is governed by how it looks as well as how it is used.

EXAM HINT

These rules are NOT given in the formula booklet. Make sure that you can use them in both directions, e.g. if you see 2^6 you can rewrite it as $(2^3)^2$ and if you see $(2^3)^2$ you can rewrite it as 2^6 . Both ways will be important!

Worked example 2.2

Evaluate $64^{\frac{2}{3}}$.

Use $a^{m/n} = \left(a^{\frac{1}{n}}\right)^m$ to split the calculation into two steps

$$\begin{aligned} 64^{\frac{2}{3}} &= \left(64^{\frac{1}{3}}\right)^2 \\ &= (4)^2 \\ \text{So, } 64^{\frac{2}{3}} &= 16 \end{aligned}$$

You must take care when expressions with *different* bases are to be combined by multiplication or division, for example $2^3 \times 4^2$. The rules such as ‘multiplication means add the exponents together’ are only true *when the bases are the same*. You cannot use this rule to simplify $2^3 \times 4^2$.

There is however, a rule that works when the bases are different *but* the exponents are the *same*.

Consider the following example:

$$\begin{aligned} 3^2 \times 5^2 &= 3 \times 3 \times 5 \times 5 \\ &= 3 \times 5 \times 3 \times 5 \\ &= 15 \times 15 \\ &= 15^2 \end{aligned}$$

This suggests the following rules:

KEY POINT 2.4

$$a^n \times b^n = (ab)^n$$

KEY POINT 2.5

$$a^n \div b^n = \left(\frac{a}{b}\right)^n$$

Exercise 2A

1. Simplify the following, leaving your answer in exponent form.

(a) (i) $6^4 \times 6^3$	(ii) $5^3 \times 5^5$
(b) (i) $a^3 \times a^5$	(ii) $x^6 \times x^3$
(c) (i) $7^{11} \times 7^{-14}$	(ii) $5^7 \times 5^{-2}$
(d) (i) $x^4 \times x^{-2}$	(ii) $x^8 \times x^{-3}$
(e) (i) $g^{-3} \times g^{-9}$	(ii) $k^{-2} \times k^{-6}$

2. Simplify the following, leaving your answer in exponent form.

(a) (i) $6^4 \div 6^3$	(ii) $5^3 \div 5^5$
(b) (i) $a^3 \div a^5$	(ii) $x^6 \div x^3$
(c) (i) $5^7 \div 5^{-2}$	(ii) $7^{11} \div 7^{-4}$
(d) (i) $x^4 \div x^{-2}$	(ii) $x^8 \div x^{-3}$
(e) (i) $2^{-5} \div 2^{-7}$	(ii) $3^{-6} \div 3^8$
(f) (i) $g^{-3} \div g^{-9}$	(ii) $k^{-2} \div k^6$

3. Express the following in the form required.

(a) (i) $(2^3)^4$ as 2^n	(ii) $(3^2)^7$ as 3^n
(b) (i) $(5^{-1})^4$ as 5^n	(ii) $(7^{-3})^2$ as 7^n
(c) (i) $(11^{-2})^{-1}$ as 11^n	
(ii) $(13^{-3})^{-5}$ as 13^n	
(d) (i) $4 \times (2^5)^3$ as 2^n	
(ii) $3^{-5} \times (9^{-1})^{-4}$ as 3^n	
(e) (i) $(4^2)^3 \times 3^{12}$ as 6^n	
(ii) $(6^3)^2 \div (2^2)^3$ as 3^n	

4. Simplify the following, leaving your answer in exponent form with a prime number as the base.

(a) (i) 4^5	(ii) 9^7
(b) (i) 8^3	(ii) 16^5
(c) (i) $4^2 \times 8^3$	(ii) $9^5 \div 27^2$
(d) (i) $4^{-3} \times 8^5$	(ii) $3^7 \div 9^{-2}$
(e) (i) $\left(\frac{1}{4}\right)^3$	(ii) $\left(\frac{1}{9}\right)^3$
(f) (i) $\left(\frac{1}{8}\right)^2 \div \left(\frac{1}{4}\right)^4$	(ii) $9^7 \times \left(\frac{1}{3}\right)^4$



5. Write the following without brackets or negative exponents:

- | | |
|---|---|
| (a) (i) $(2x^2)^3$ | (ii) $(3x^4)^2$ |
| (b) (i) $2(x^2)^3$ | (ii) $3(x^4)^2$ |
| (c) (i) $\frac{(3a^3)^4}{9a^2}$ | (ii) $\frac{(4x)^4}{8(2x)^4}$ |
| (d) (i) $(2x)^{-1}$ | (ii) $\left(\frac{3}{y}\right)^{-2}$ |
| (e) (i) $2x^{-1}$ | (ii) $\frac{3}{y^{-2}}$ |
| (f) (i) $5 \div \left(\frac{3}{xy^2}\right)^2$ | (ii) $\left(\frac{ab}{2}\right)^3 \div \left(\frac{a}{b}\right)^2$ |
| (g) (i) $\left(\frac{2}{q}\right)^2 \div \left(\frac{p}{2}\right)^{-3}$ | (ii) $\left(\frac{6}{x}\right)^4 \div \left(2 \times \frac{3^2}{x}\right)^{-3}$ |



6. Simplify the following:

- | | |
|--|--|
| (a) (i) $(x^6)^{\frac{1}{2}}$ | (ii) $(x^9)^{\frac{4}{3}}$ |
| (b) (i) $(4x^{10})^{0.5}$ | (ii) $(8x^{12})^{-\frac{1}{3}}$ |
| (c) (i) $\left(\frac{27x^9}{64}\right)^{-\frac{1}{3}}$ | (ii) $\left(\frac{x^4}{y^8}\right)^{-1.5}$ |



7. Solve for x , giving your answer as a rational value:

- | | |
|------------------------------------|-------------------------------------|
| (a) (i) $8^x = 32$ | (ii) $25^x = \frac{1}{125}$ |
| (b) (i) $\frac{1}{49^x} = 7$ | (ii) $\frac{1}{16^x} = 8$ |
| (c) (i) $2 \times 3^x = 162$ | (ii) $3 \div 5^x = 0.12$ |
| (d) (i) $2 \times 5^{x-1} = 250$ | (ii) $5 + 3^{x+2} = 14$ |
| (e) (i) $16 + 2^x = 2^{x+1}$ | (ii) $100^{x+5} = 10^{3x-1}$ |
| (f) (i) $6^{x+1} = 162 \times 2^x$ | (ii) $4^{1.5x} = 2 \times 16^{x-1}$ |

In Section 2G you will see that there is an easier way to solve equations like this when you have a calculator and can use logarithms.



8. Any simple computer program is able to sort n input values in $k \times n^{1.5}$ microseconds. Observations show that it sorts a million values in half a second. Find the value of k . [3 marks]



9. A square-ended cuboid has volume xy^2 , where x and y are lengths. A cuboid for which $x = 2y$ has volume 128 cm^3 . Find x . [3 marks]

π and e have many similar properties. Both are irrational, meaning that they cannot be written as a ratio of two whole numbers and both are transcendental, meaning that they cannot be written as the solution to a polynomial equation. The proof of these facts is intricate but beautiful.



In chapter 16 you will see that e plays a major role in studying rates of change.



In fact, as can be seen by taking greater and greater values of n , the resultant increase factor $\left(1 + \frac{1}{n}\right)^n$ tends towards a value of approximately 2.71828182849 which, much like π , is such an important part of mathematical studies that it has been given its own letter, e .

KEY POINT 2.7

$$e = 2.71828182849\dots$$

Although e has many important properties in mathematics, it is still just a number, and so all the standard rules of arithmetic and exponents apply. You are not likely to find an exam question just on e , but the number e is likely to appear in questions on many other topics.

EXAM HINT

In questions involving the number e you may be asked to either give an exact answer (such as e^2) or to use your calculator, in which case you should usually round the answer to 3 significant figures.

Exercise 2C



1. Find the values of the following to 3 significant figures:

- | | |
|--------------------|----------------------|
| (a) (i) $e + 1$ | (ii) $e - 4$ |
| (b) (i) $3e$ | (ii) $\frac{e}{2}$ |
| (c) (i) e^2 | (ii) e^{-3} |
| (d) (i) $5e^{0.5}$ | (ii) $\frac{3}{e^7}$ |

2. Evaluate $\sqrt[6]{(\pi^4 + \pi^5)}$. What do you notice about this result?

See Supplementary sheet 2 'Logarithmic scales and log-log graphs' on the CD-ROM if you would like to discover more about logarithms for yourself.



2D Introducing logarithms

In this section we shall look at a new operation which reverses the effect of exponentiating (that is, raising a number to an exponent).

If you are asked to solve

$$x^2 = 3, x \geq 0$$

you can either find a decimal approximation (using a calculator or trial and improvement) or you can use the square root symbol: $x = \sqrt{3}$.

This restates the problem as 'x is the positive value which when squared gives 3'.

Similarly, if asked to solve:

$$10^x = 50$$

you could use trial and improvement to seek a decimal value:

$$10^1 = 10$$

$$10^2 = 100$$

So x is between 1 and 2:

$$10^{1.5} = 31.6$$

$$10^{1.6} = 39.8$$

$$10^{1.7} = 50.1$$

So the answer is around 1.7.

However, just as with squares and square roots there is also a function to answer the question 'What is the number, which when put as the exponent of 10, gives this value?'

The function is called a base-10 **logarithm**, written $\log_{10}$.

So in the above example: $10^x = 50$ so $x = \log_{10} 50$.

This means that $y = 10^x$ may be re-expressed as $x = \log_{10}(y)$.

In fact, the base involved need not be 10, but could be any positive value other than 1.

KEY POINT 2.8

$$b = a^x \Leftrightarrow x = \log_a b$$



It is worth noting that the two most common bases have abbreviations for their logarithms.

Since we use a decimal system of counting, base 10 is the default base for a logarithm, so that $\log_{10} x$ is generally written more simply as just $\log x$, called the **common logarithm**. Also e, encountered in Section 2C, is considered the 'natural' base, and its counterpart the **natural logarithm** is denoted by $\ln x$.

KEY POINT 2.9

$\log_{10} x$ is often written as $\log x$.

$\log_e x$ is often written as $\ln x$.



The symbol $\Leftrightarrow$ has a very specific meaning (implies and is implied by) in mathematical logic. It means that if the left-hand side is true then so is the right-hand side, and also if the right-hand side is true then so is the left-hand side.

This is the form written in the Formula booklet; you only need to remember that it means you can switch between them.

➤ See Key point 2.18.



Since taking a logarithm reverses the process of exponentiation, it follows that:

KEY POINT 2.10

$$\log_a(a^x) = x$$

KEY POINT 2.11

$$a^{\log_a x} = x$$

These are sometimes referred to as the cancellation principles. This sort of 'cancellation', similar to stating that (for positive x) $\sqrt[n]{x^n} = x = \left(\sqrt[n]{x}\right)^n$, is frequently useful when simplifying logarithm expressions, but you can only apply it when the base of the logarithm and the base of the exponential match.

Worked example 2.6

 Evaluate:

- (a) $\log_5 625$ (b) $\log_8 16$

Express the argument of the logarithm in exponent form with the same base

Apply the cancellation principle $\log_a(a^x) = x$

16 is not a power of 8, but they are both powers of 2

We need to convert 2^4 to an exponent of 8 = 2^3

Rewrite 4 as $3 \times \frac{4}{3}$ and use $a^{mn} = (a^m)^n$

Apply the cancellation principle $\log_a(a^x) = x$

$$(a) \log_5 625 = \log_5 5^4$$

$$= 4$$

$$(b) \log_8(16) = \log_8(2^4)$$

$$= \log_8\left(2^{3 \times \frac{4}{3}}\right) = \log_8\left(8^{\frac{4}{3}}\right)$$

$$= \frac{4}{3}$$

Actually, you can find the logarithm of a negative number, but the answer turns

 out not to be real; it  is a complex number, which is a new type of number you will meet in chapter 15.

Whenever you raise a positive number to a power, positive or negative, the answer is always positive. Therefore we currently have no answer to a question such as $10^x = -3$.

This means:

KEY POINT 2.12

The logarithm of a negative number or zero has no real value.

Exercise 2D

1. Evaluate the following:
- | | |
|------------------------------|----------------------------|
| (a) (i) $\log_3 27$ | (ii) $\log_4 16$ |
| (b) (i) $\log_5 5$ | (ii) $\log_3 3$ |
| (c) (i) $\log_{12} 1$ | (ii) $\log_{15} 1$ |
| (d) (i) $\log_3 \frac{1}{3}$ | (ii) $\log_4 \frac{1}{64}$ |
| (e) (i) $\log_4 2$ | (ii) $\log_{27} 3$ |
| (f) (i) $\log_8 \sqrt{8}$ | (ii) $\log_2 \sqrt{2}$ |
| (g) (i) $\log_8 4$ | (ii) $\log_{81} 27$ |
| (h) (i) $\log_{25} 125$ | (ii) $\log_{16} 32$ |
| (i) (i) $\log_4 2\sqrt{2}$ | (ii) $\log_9 81\sqrt{3}$ |
| (j) (i) $\log_{25} 0.2$ | (ii) $\log_4 0.5$ |
2. Use a calculator to evaluate each of the following, giving your answer correct to 3 significant figures:
- | | |
|-------------------|-------------------------------------|
| (a) (i) $\log 50$ | (ii) $\log\left(\frac{1}{4}\right)$ |
| (b) (i) $\ln 0.1$ | (ii) $\ln 10$ |
3. Simplify the following expressions:
- | | |
|---|--|
| (a) (i) $7\log x - 2\log x$ | (ii) $2\log x + 3\log x$ |
| (b) (i) $(\log x - 1)(\log y + 3)$ | (ii) $(\log x + 2)^2$ |
| (c) (i) $\frac{\log a + \log b}{\log a \log b}$ | (ii) $\frac{(\log a)^2 - 1}{\log a - 1}$ |
4. Make x the subject of the following:
- | | |
|----------------------------|-----------------------|
| (a) (i) $\log_3 x = y$ | (ii) $\log_4 x = 2y$ |
| (b) (i) $\log_a x = 1 + y$ | (ii) $\log_a x = y^2$ |
| (c) (i) $\log_x 3y = 3$ | (ii) $\log_x y = 2$ |
5. Find the value of x in each of the following:
- | | |
|--------------------------|--------------------------------|
| (a) (i) $\log_2 x = 32$ | (ii) $\log_2 x = 4$ |
| (b) (i) $\log_5 25 = 5x$ | (ii) $\log_{49} 7 = 2x$ |
| (c) (i) $\log_x 36 = 2$ | (ii) $\log_x 10 = \frac{1}{2}$ |

EXAM HINT

On most calculators $\log x$ is denoted with a $\boxed{\log}$ button, and $\ln x$ by a $\boxed{\ln}$ button.

EXAM HINT

Remember that ' $\log x$ ' is just a number so can be treated the same way as any variable.



6. Solve the equation $\log_{10}(9x+1) = 3$. [4 marks]

7. Solve the equation $\log_8 \sqrt{1-x} = \frac{1}{3}$. [4 marks]

8. Find the exact solution to the equation $\ln(3x-1) = 2$. [5 marks]



9. Find all values of x which satisfy $(\log_3 x)^2 = 4$. [5 marks]



10. Solve the simultaneous equations:

$$\log_3 x + \log_5 y = 6$$

$$\log_3 x - \log_5 y = 2$$
 [6 marks]

11. Solve the equation $3(1 + \log x) = 6 + \log x$. [5 marks]

12. Solve the equation $\log_x 4 = 9$. [4 marks]

13. The Richter scale is a way of measuring the strength of earthquakes. An increase of one unit on the Richter scale corresponds to an increase by a factor of 10 in the strength of the earthquake. What would be the Richter level of an earthquake which is twice as strong as a level 5.2 earthquake? [5 marks]

2E Laws of logarithms



Just as there are rules which hold when working with exponents, so there are corresponding rules which apply to logarithms. These are derived from the laws of exponents and can be found on the Fill-in proof sheet 18. 'Differentiating logarithmic functions graphically' on the CD-ROM.

- The logarithm of a *product* is the *sum* of the logarithms.

KEY POINT 2.13

$$\log_a xy = \log_a x + \log_a y \text{ for } x, y > 0$$

- The logarithm of a *quotient* is the *difference* of the logarithms.

KEY POINT 2.14

$$\log_a \frac{x}{y} = \log_a x - \log_a y \text{ for } x, y > 0$$

EXAM HINT

These formulae are not in the Formula booklet.

- The logarithm of a *reciprocal* is the *negative* of the logarithm.

KEY POINT 2.15

$$\log_a \frac{1}{x} = -\log_a x \text{ for } x > 0$$

EXAM HINT

It is important to know what you *cannot* do with logarithms. One very common mistake is to try to simplify $\log(x + y)$ into either $\log x + \log y$ or $\log x \log y$.

- The logarithm of a *exponent* is the *multiple* of the logarithm.

KEY POINT 2.16

$$\log_a x^p = p \log_a x \text{ for } x > 0$$

- The logarithm of 1 is always 0, irrespective of the base.

KEY POINT 2.17

$$\log_a 1 = 0$$

The rules of logarithms can be used to manipulate expressions and solve equations involving logarithms.

Worked example 2.7

If $x = \log_{10} a$ and $y = \log_{10} b$, express $\log_{10} \frac{100a^2}{b}$ in terms of x , y and integers.

Use laws of logs to isolate $\log_{10} a$ and $\log_{10} b$

First, use $\log \frac{x}{y} = \log x - \log y$

...then $\log xy = \log x + \log y$...

... then $\log x^p = p \log x$...

... then calculate $\log_{10} 100$

... then write in terms of x and y

$$\log_{10} \frac{100a^2}{b} = \log_{10} (100a^2) - \log_{10} b$$

$$= \log_{10} 100 + \log_{10} a^2 - \log_{10} b$$

$$= \log_{10} 100 + 2\log_{10} a - \log_{10} b$$

$$= 2 + 2\log_{10} a - \log_{10} b$$

$$= 2 + 2x - y$$

Worked example 2.8

Solve the equation $\log_2 x + \log_2 (x + 4) = 5$.

Rewrite one side as a single logarithm
using $\log x + \log y = \log xy$

Undo the logarithm by exponentiating
both sides with base 2 (to balance, you
must also exponentiate 5 by base 2) ...

and use the cancellation principle

Use standard methods for quadratic
equations

Check your solution in the original
equation

$$\log_2 x + \log_2 (x + 4) = 5$$

$$\Leftrightarrow \log_2 (x(x + 4)) = 5$$

$$\Leftrightarrow 2^{\log_2 (x(x+4))} = 2^5$$

$$\Leftrightarrow x^2 + 4x = 32$$

$$\Leftrightarrow x^2 + 4x - 32 = 0$$

$$\Leftrightarrow (x + 8)(x - 4) = 0$$

$$\Rightarrow x = -8 \text{ or } x = 4$$

When $x = -8$:

LHS: $\log_2(-8)$ and $\log_2(-4)$ are not real so
this solution does not work

When $x = 4$:

$$\begin{aligned} \text{LHS: } \log_2 4 + \log_2 8 &= 2 + 3 = 5 \\ &= \text{RHS} \end{aligned}$$

EXAM HINT

Checking your solutions is more than just looking for an arithmetic error – as you can see from the example above, false solutions can occur through *correct* algebraic manipulation.

You will notice that although we have discussed logarithms for a general base, a , your calculator may only have buttons for the common logarithm and the natural logarithm ($\log x$ and $\ln x$).

To use a calculator to evaluate, for example, $\log_5 20$ you can use the **change of base rule** of logarithms.

KEY POINT 2.18

Change of base rule for logarithms:

$$\log_b a = \frac{\log_c a}{\log_c b}$$



So, we can calculate $\log_5 20$ using the logarithm to the base 10:

$$\log_5 20 = \frac{\log 20}{\log 5} = 1.86 \text{ (3 SF)}$$

The change of base rule is useful for more than just evaluating logarithms.



For an insight into what mathematics was like before calculators, have a look at Supplementary sheet 3, 'A history of logarithms' on the CD-ROM.



Worked example 2.9

Solve the equation $\log_3 x + \log_9 x = 2$.

We want to have logarithms involving just one base, so use the change of base rule to turn logs with base 9 into logs with base 3

Collecting the logs together

Exponentiate both sides with base 3

$$\log_9 x = \frac{\log_3 x}{\log_3 9} = \frac{\log_3 x}{2}$$

Therefore:

$$\log_3 x + \log_9 x = 2$$

$$\Leftrightarrow \log_3 x + \frac{\log_3 x}{2} = 2$$

$$\Leftrightarrow \frac{3}{2} \log_3 x = 2$$

$$\Leftrightarrow \log_3 x = \frac{4}{3}$$

$$\Leftrightarrow x = 3^{\frac{4}{3}} = 4.33 \text{ (3SF)}$$

Equations combining logarithms and quadratics can be found in Section 4B.

Exercise 2E

- Given $b > 0$, simplify each of the following:
 - (i) $\log_b b^4$ (ii) $\log_b \sqrt{b}$
 - (i) $\log_{\sqrt{b}} b^3$ (ii) $\log_b b^2 - \log_{b^2} b$
- If $x = \log a$, $y = \log b$ and $z = \log c$, express the following in terms of x , y and z :
 - (i) $\log b^7$ (ii) $\log a^2 b$
 - (i) $\log\left(\frac{ab^2}{c}\right)$ (ii) $\log\left(\frac{a^2}{bc^3}\right)$

- (c) (i) $\log\left(\frac{100}{bc^5}\right)$ (ii) $\log(5b) + \log(2c^2)$
 (d) (i) $\log a^3 - 2\log ab^2$ (ii) $\log(4b) + 2\log(5ac)$
 (e) (i) $\log_a a^2b$ (ii) $\log_b\left(\frac{a}{bc}\right)$
 (f) (i) $\log_{a^b}(b^a)$ (ii) $\log_{ab} ac^2$



3. Solve the following for x :

- (a) (i) $\log_3\left(\frac{2+x}{2-x}\right) = 3$ (ii) $\log_2(7x+4) = 5$
 (b) (i) $\log_3 x - \log_3(x-6) = 1$ (ii) $\log_8 x - 2\log_8\left(\frac{1}{x}\right) = 1$
 (c) (i) $\log_2 x + 1 = \log_4 x$ (ii) $\log_8 x + \log_2 x = 4$
 (d) (i) $\log_4 x + \log_8 x = 2$ (ii) $\log_{16} x - \log_{32} x = 0.5$
 (e) (i) $\log_3(x-7) + \log_3(x+1) = 2$
 (ii) $2\log(x-2) - \log x = 0$
 (f) (i) $\log(x^2+1) = 1 + 2\log x$
 (ii) $\log(3x+6) = \log 3 + 1$

4. Find the exact solution to the equation $2\ln x + \ln 9 = 3$, giving your answer in the form Ae^B where A and B are rational numbers. [5 marks]



5. If $a = \ln 2$ and $b = \ln 5$, find in terms of a and b : [6 marks]

- (a) $\ln 50$ (b) $\ln 0.16$

6. Solve $\log_2 x = \log_x 2$. [5 marks]

7. Prove that if $a^x = b^y = (ab)^{xy}$ where $a, b > 1$ then $x + y = 1$ or $x = y = 0$. [5 marks]



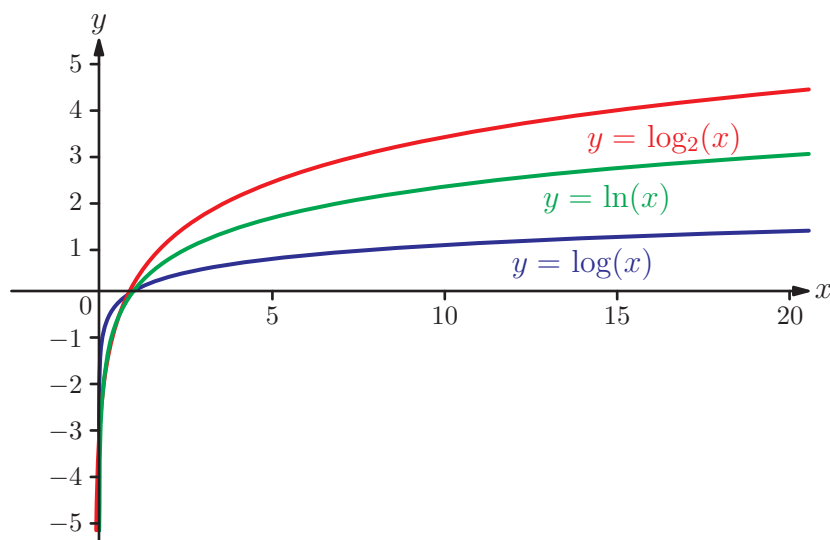
8. Evaluate $\log \frac{1}{2} + \log \frac{2}{3} + \log \frac{3}{4} + \log \frac{4}{5} \dots + \log \frac{8}{9} + \log \frac{9}{10}$. [4 marks]

9. Given that $\log_a b = \log_b a$, and that $a, b \neq 1$ and $a \neq b$, find b in terms of a . [5 marks]

2F Graphs of logarithms

It is important also to know the graph of the logarithm function, and the various properties of logarithms which we can deduce from it.

Below are the graphs of $y = \log x$, $y = \log_2 x$ and $y = \ln x$:



These three curves all have a similar shape.

The change of base rule (Key point 2.18) states that $\log_b a = \frac{\log_c a}{\log_c b}$.

So, $\log_2 x = \frac{\log x}{\log 2}$ and $\ln x = \frac{\log x}{\log e}$ and all other logarithm functions are simply multiples of the common logarithm function $y = \log x$.

The graph shows the following important facts about the logarithm function (Key point 2.19).

KEY POINT 2.19

If $y = \log_a x$ then:

- The graph of y against x crosses the x -axis at $(1, 0)$, because $\log_a 1 = 0$ (for any positive value of a).
- $\log x$ is negative for $0 < x < 1$ and positive for $x > 1$.
- The graph of the function lies entirely to the right of the y -axis, since the logarithm of a negative value does not produce a real solution.
- The logarithm graph increases throughout: as x tends to infinity so does y .
- The y -axis is an asymptote to the curve.

In chapter 6 you will see how this type

of change causes a vertical stretch of the graph.

EXAM HINT

Vertical asymptotes are even harder to detect accurately from a calculator display than the horizontal ones. This is why it is important that you know how to find an asymptote of a logarithmic graph.

You may observe that a logarithm graph is the reflection

of an exponential graph. You will see why this is the case in chapter 5.

Exercise 2F

It is unlikely you will find exam questions testing *just* this topic, but you may be required to sketch a graph involving a logarithm as a part of another question.

1. Sketch the following graphs, labelling clearly the vertical asymptote and all axis intercepts.

(a) (i) $y = \log(x^2)$ (ii) $y = \log(x^3)$

(b) (i) $y = \log 4x$ (ii) $y = \log 2x$

(c) (i) $y = \log(x-2)$ (ii) $y = \log(x+1)$

2. Why are the graphs of $y = \log(x^2)$ and $y = 2\log(x)$ different?

2G Solving exponential equations

One of the main uses of logarithms is to solve equations with the unknown in the exponent. By taking logarithms, the unknown becomes a factor, which is easier to deal with.

Worked example 2.10

Solve the equation $3 \times 2^x = 5^{x-1}$, giving your answer in the form $\frac{\log p}{\log q}$ where p and q are rational numbers.

Since the unknown is in the exponent, taking logarithms is a good idea

Use the rules of logarithms to simplify the expression

Expand the brackets and get all of the x 's on one side and everything else on the other side

Factorise and divide to find x

Use the rules of logarithms to write it in the correct form

$$\log(3 \times 2^x) = \log(5^{x-1})$$

$$\log 3 + \log 2^x = \log(5^{x-1})$$

$$\log 3 + x \log 2 = (x-1) \log 5$$

$$\log 3 + x \log 2 = x \log 5 - \log 5$$

$$\log 3 + \log 5 = x \log 5 - x \log 2$$

$$x = \frac{\log 3 + \log 5}{\log 5 - \log 2}$$

$$x = \frac{\log 15}{\log\left(\frac{5}{2}\right)}$$

EXAM HINT

A common mistake is to say that $\log(3 \times 2^x)$ is $\log 3 \times \log 2^x$. Make sure you learn the rules of logarithms carefully.



Logarithmic scales make it easy to compare values which are very different. Some of these scales are explored in Supplementary sheet 2 on the CD-ROM.



You can use similar ideas to solve inequalities with the unknown in the exponent:

Worked example 2.11

The number of bacteria in a culture medium is given by $N = 1000 \times 2^{4t}$, where t is the number of hours elapsed since 08:00. At what time will the population first exceed one million?

Simplify the equation where possible

$$1000 \times 2^{4t} \geq 1\,000\,000 \\ \Leftrightarrow 2^{4t} \geq 1000$$

Take logarithms of each side

$$\Leftrightarrow \log(2^{4t}) \geq \log 1000$$

Use the rule $\log_a x^p = p \log_a x$

$$\Leftrightarrow 4t \log 2 \geq 3$$

Note that $\log 2 > 0$, so the inequality remains in the same orientation when we divide

$$\Leftrightarrow t \geq \frac{3}{4 \log 2} = 2.49(35\text{F})$$

To answer the question, convert 2.49 to hours and minutes, then add it to 08:00

10:29



You know that you can always apply the same operation to both sides of an equation, but in this worked example we took logarithms of both sides of an inequality. You might like to investigate for which operations this is valid.

EXAM HINT

Whenever dividing an inequality by a logarithm it is important to remember to check if it is positive or negative.



There is another type of exponential equation which you will meet – a disguised quadratic equation. These are explored in Section 4B.



Exercise 2G

1. Solve for x , giving your answer correct to 3 significant figures.
 - (a) (i) $3 \times 4^x = 90$ (ii) $1000 \times 1.02^x = 10\,000$
 - (b) (i) $6 \times 7^{3x+1} = 1.2$ (ii) $5 \times 2^{2x-5} = 94$
 - (c) (i) $3^{2x} = 4^{x-1}$ (ii) $5^x = 6^{1-x}$
 - (d) (i) $3 \times 2^{3x} = 7 \times 3^{3x-2}$ (ii) $4 \times 8^{x-1} = 3 \times 5^{2x+1}$
2. In a yeast culture, cell numbers are given by $N = 100e^{1.03t}$, where t is measured in hours after the cells are introduced to the culture.
 - (a) What is the initial number of cells? [1 mark]
 - (b) How many cells will be present after 6 hours? [1 mark]
 - (c) How long will it take for the population to exceed one thousand? [2 marks]
3. A rumour spreads exponentially through a school. When school begins (at 9 a.m.) 18 people know it. By 10 a.m. 42 people know it.
 - (a) How many people know it at 10.30? [3 marks]
 - (b) There are 1200 people in the school. According to the exponential model, at what time will everyone know the rumour? [2 marks]
4. In an experimental laboratory, a scientist sets up a positive feedback loop for a fission reaction and extracts heat to control the experiment and produce power. When the reaction is established, and while sufficient fuel is present, the power he can siphon off is given by $P = 32(e^{0.0012t} - 1)$, where P is measured in units of energy per second and t in seconds.
 - (a) How much energy is being produced after 2 minutes? [1 mark]
 - (b) The equipment reaches a dangerous temperature when P exceeds 7×10^5 . For how long can the experiment safely be run? [2 marks]
5. The weight of a block of salt W in a salt solution after t seconds is given by:
$$W = ke^{-0.01t}$$
 - (a) Sketch the graph of W against t . [2 marks]
 - (b) How long will it take to reach 25% of its original weight? [2 marks]

6. Solve the equation $5 \times 4^{x-1} = \frac{1}{3^{2x}}$, giving your answer in the form $x = \frac{\ln p}{\ln q}$, where p and q are rational numbers. [5 marks]
7. Solve the equation $\frac{1}{7^x} = 3 \times 49^{5-x}$, giving your answer in the form $a + \log_7 b$ where $a, b \in \mathbb{Z}$.
8. A cup of tea is poured at 98°C . After two minutes it has reached 94°C . The difference between the temperature of the tea and the room temperature (22°C) falls exponentially. Find the time it takes for the tea to cool to 78°C . [5 marks]
9. (a) Show that the equation $3^x = 3 - x$ has only one solution. [2 marks]
 (b) Find the solution, giving your answer to 3 significant figures. [4 marks]

Summary

- In this chapter, we revisited the rules for **exponents**.

$$a^m \times a^n = a^{m+n}$$

$$a^m \div a^n = a^{m-n}$$

$$a^n \times b^n = (ab)^n$$

$$a^n \div b^n = \left(\frac{a}{b}\right)^n$$

$$(a^m)^n = a^{m \times n}$$

$$a^{\frac{m}{n}} = \left(\sqrt[n]{a}\right)^m = \sqrt[n]{a^m}$$

- Exponential equations can be used to model growth and decay of some simple real-life systems, taking as a general form the function:

$$N = Ba^{\left(\frac{t}{k}\right)} + c$$

- $\log_a b$ asks the question: 'What exponent do I have to raise a to in order to get b ?'

$$b = a^x \Leftrightarrow x = \log_a b$$

- e is the mathematical constant, (Euler's number): $e = 2.71828182849\dots$
- $\log_{10} x$ (common logarithm) is often written as $\log x$.

- $\log_e x$ (natural logarithm) is often written as $\ln x$.
- **Logarithms** undo the effect of exponentiating and vice versa:

$$\log_a(a^x) = x \Leftrightarrow x = a^{\log_a x}$$

- Logarithms obey these rules, which are *not* covered in the Formula booklet.

$$\log_a xy = \log_a x + \log_a y$$

$$\log_a \left(\frac{x}{y} \right) = \log_a x - \log_a y$$

$$\log_a \left(\frac{1}{x} \right) = -\log_a x$$

$$\log_a x^p = p \log_a x$$

$$\log_a 1 = 0$$

- Taking the logarithm of a negative number or zero does not give a real value.
- There is also the change of base formula (not in the Formula booklet): $\log_b a = \frac{\log_c a}{\log_c b}$ and a related rule for exponents (*is* in the Formula booklet): $a^x = e^{x \ln a}$
- The following properties of the logarithm function can be deduced from its graph, if $y = \log_a x$ then:
 - The graph of y against x crosses the x -axis at $(1,0)$
 - $\log x$ is negative for $0 < x < 1$ and positive for $x > 1$
 - The graph lies to the right of the y -axis; the y -axis is an asymptote to the curve
 - The logarithm graph increases throughout; as x tends to infinity so does y .
- Logarithms are used to solve many exponential equations.

Introductory problem revisited

A radioactive substance has a half-life of 72 years. A 1 kg block of the substance is found to have a radioactivity of 25 million Becquerel (Bq). How long, to the nearest 10 years, would it take for the radioactivity to have fallen below 10 000 Bq?

Establish the exponential equation

Initial condition gives B

Every time t increases by 72,
 R falls by 50%

Let R be the radioactivity after t years

$$R = Ba^{\frac{t}{k}}$$

$$\text{When } t = 0, R = 25 \times 10^6 = B$$

$$a = 0.5, k = 72$$

$$C = 25 \times 10^6 \times 0.5^{\frac{t}{72}}$$

continued...

The unknown is in the exponent so take
logarithms

Now rearrange to find t
Note that $\log(0.5) < 0$, so the inequality is
reversed when dividing through

$$\text{Require } R \leq 10^4$$

$$25 \times 10^6 \times 0.5^{\frac{t}{72}} \leq 10^4$$

$$\Leftrightarrow 0.5^{\frac{t}{72}} \leq 0.0004$$

$$\Leftrightarrow \log\left(0.5^{\frac{t}{72}}\right) \leq \log(0.0004)$$

$$\frac{t}{72} \log(0.5) \leq \log(0.0004)$$

$$\Leftrightarrow t \geq \frac{72 \log(0.0004)}{\log(0.5)} = 812.7$$

It takes around 810 years for the
radioactivity to fall below 10 000 Bq.